Sleep Health and Lifestyle Study

Team: SnoozeStats

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01

INTRODUCTION & BACKGROUND

Dataset Information, Analysis Goals, & Data Cleaning

Data Set Information

Observations

400 individuals

Variables

12 columns (excluding Person ID)
Demographics, sleep metrics, lifestyle habits, cardiovascular measures

Data

Comprehensive metrics on sleep, lifestyle, & cardiovascular health

Response Variable

Sleep Duration (hours)

Predictor Variables

Gender, Age, Occupation, Physical Activity (minutes/day), Stress Level (scale: 1-10), BMI Category, Blood Pressure (systolic/diastolic), Heart Rate (bpm), Daily Steps

Analysis Goals

Problem

Understanding factors influencing sleep duration

Goal

To determine the best predictors for sleep duration using multiple linear regression models

Solution

Apply exploratory data analysis and variable selection to help identify key influences on sleep health

Data Cleaning

Removing Columns

Removed `Person.id` — no contribution towards analysis

Separating Blood Pressure

Split up BP by "/" into two columns: `Systolic` & `Diastolic`

Standardizing BMI Categories

Replaced instances of "Normal Weight" in `BMI_Category` with "Normal"

Converting Data Types

Turned categorical variables into factors and integers to numeric values

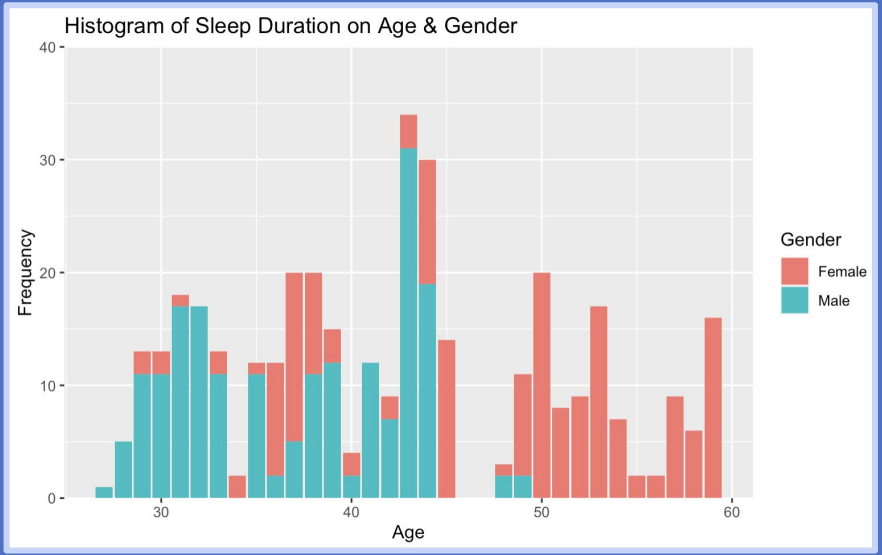
```
> str(sleep)
'data.frame':   374 obs. of  13 variables:
 $ Gender           : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
 $ Age              : num  27 28 28 28 28 28 29 29 29 29 ...
 $ Occupation       : Factor w/ 11 levels "Accountant","Doctor",..: 10 2 2 7 7 10 11 2 2 2 ...
 $ Sleep_Duration   : num  6.1 6.2 6.2 5.9 5.9 5.9 6.3 7.8 7.8 7.8 ...
 $ Sleep_Quality    : num  6 6 6 4 4 4 6 7 7 7 ...
 $ Physical_Activity: num  42 60 60 30 30 30 40 75 75 75 ...
 $ Stress_Level     : num  6 8 8 8 8 8 7 6 6 6 ...
 $ BMI_Category     : Factor w/ 3 levels "Normal","Obese",..: 3 1 1 2 2 2 2 1 1 1 ...
 $ SystolicBP       : num  126 125 125 140 140 140 140 120 120 120 ...
 $ DiastolicBP      : num  83 80 80 90 90 90 90 80 80 80 ...
 $ Heart_Rate       : num  77 75 75 85 85 85 82 70 70 70 ...
 $ Daily_Steps      : num  4200 10000 10000 3000 3000 3000 3500 8000 8000 8000 ...
 $ Sleep_Disorder   : Factor w/ 3 levels "Insomnia","None",..: 2 2 2 3 3 1 1 2 2 2 ...
```

02

EXPLORATORY ANALYSIS

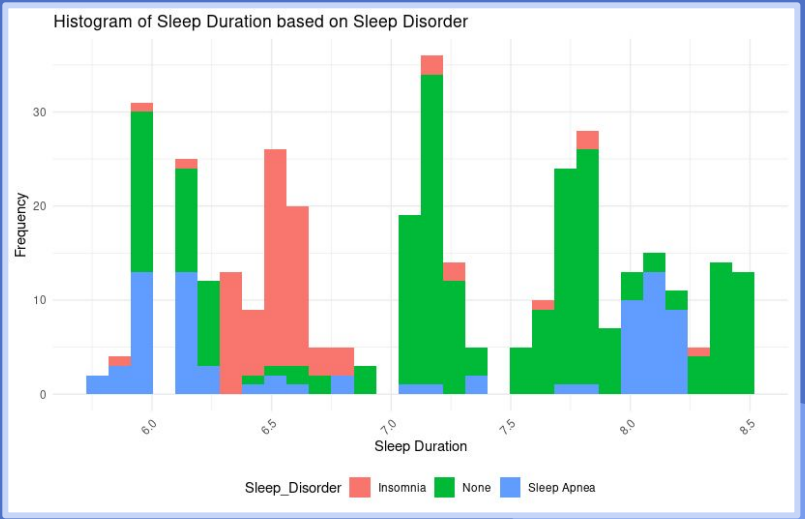
Scatter plots, bar plots, etc

Exploratory Data Analysis

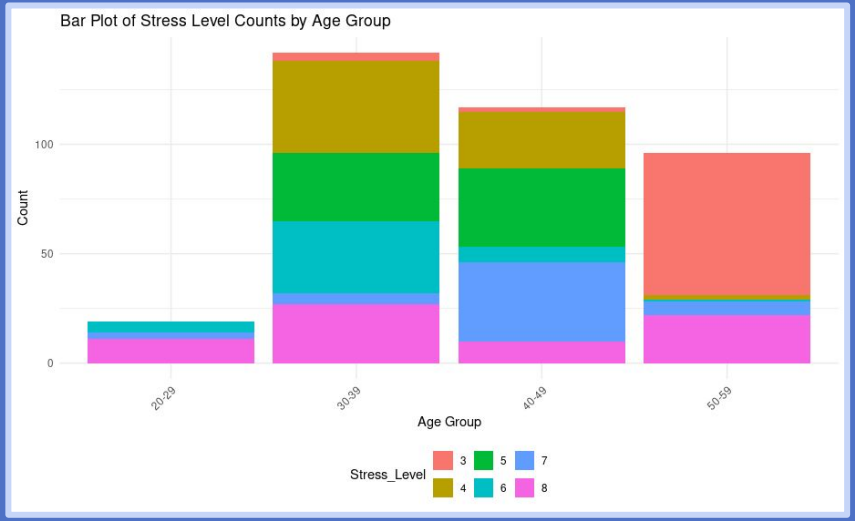


- The younger age groups are majorly comprised of males.
- In contrast, females predominantly represent the older age categories.

- Insomnia is characterized by a reduced number of sleeping hours.
- Sleep Apnea can occur during any period of sleep, regardless of duration

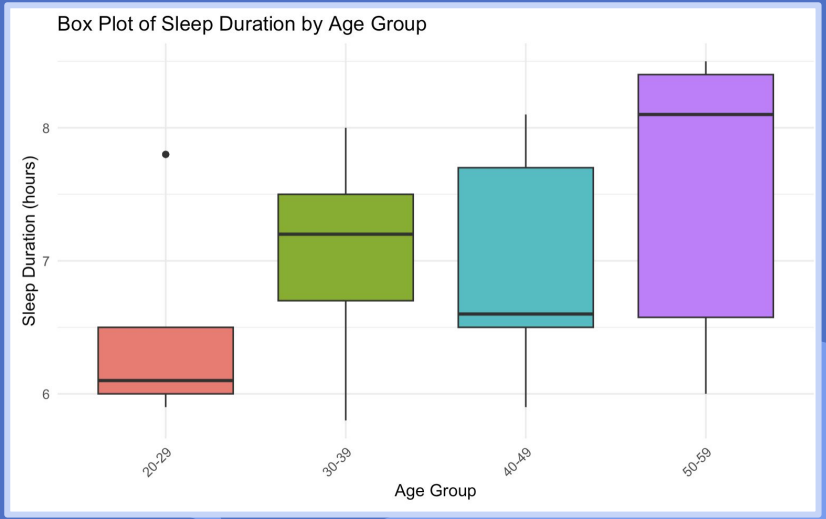


Exploratory Data Analysis Continued

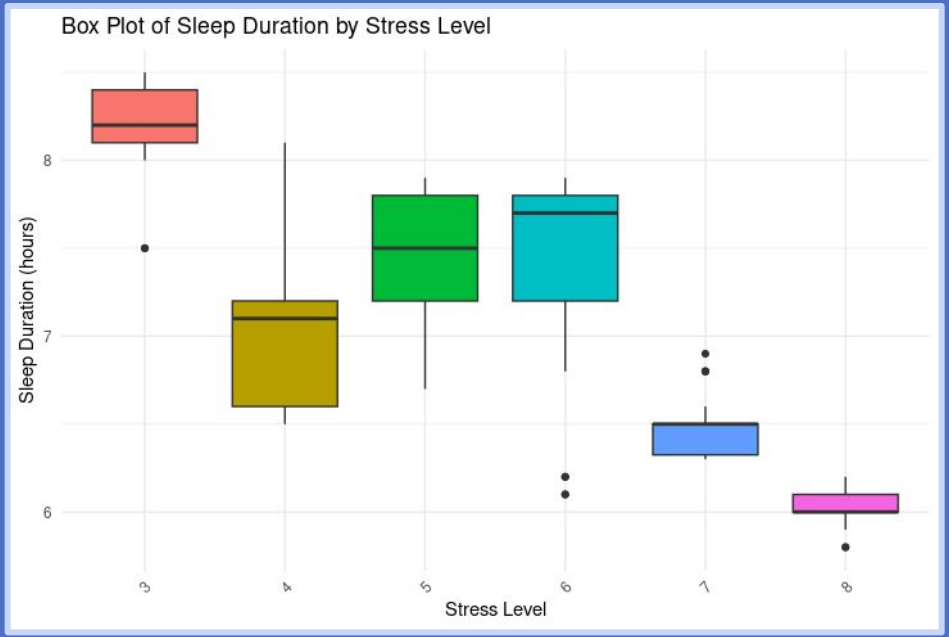


- Stress levels are not uniform across age groups.
- Individuals aged 50 to 59 often report lower stress levels.
- This trend contrasts with other age cohorts who generally experience higher stress.

- Age 20-29: Average sleep duration is 6 hours.
- Age 30-39: Sleep duration typically falls between 7-8 hours.
- Age 40-49: The majority sleep about 5 hours.
- Age 50-59: Sleep duration varies, ranging from 6 to 9 hours with an average of 8 hours.

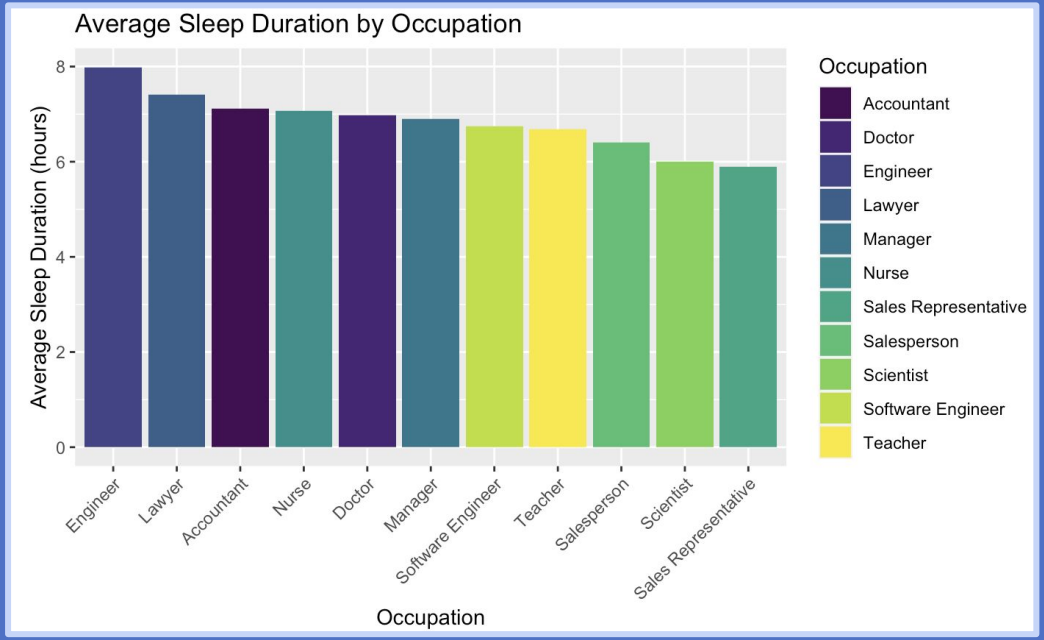


Exploratory Data Analysis Continued



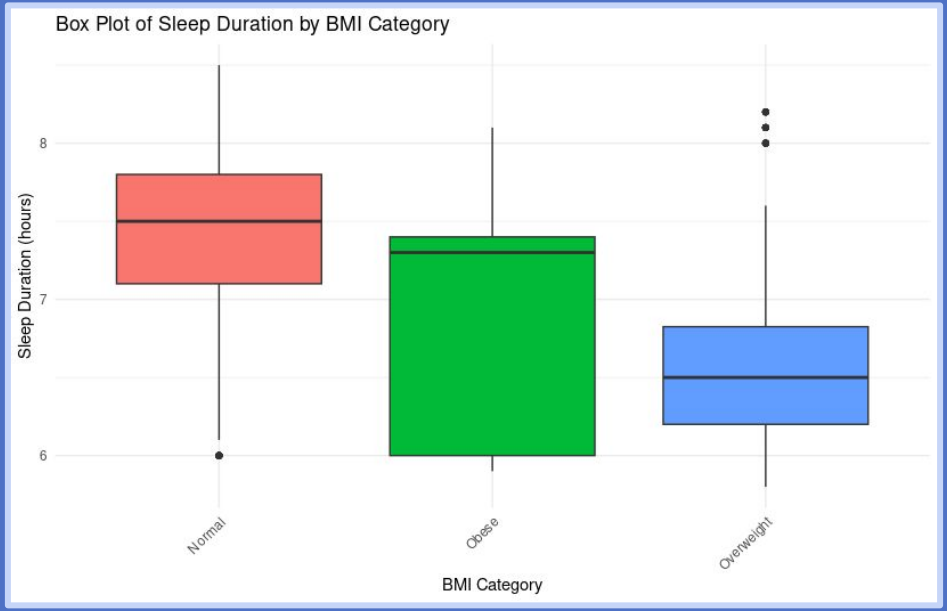
- Stress levels have an impact on sleep duration.
- The connection is mild for stress levels rated between 4 to 6.

Exploratory Data Analysis Continued



- Engineers record the highest average sleep duration, while Sales Representatives experience the least average amount of sleep.

Exploratory Data Analysis Continued



- Individuals with a normal weight typically average 7.5 hours of sleep.
- Those who are overweight tend to sleep fewer hours compared to individuals of normal weight.

03

VARIABLE SELECTION & Residual Analysis

Full Model & Reduced Model

Full Model

Full Model with all 12 predictors

```
Call:
lm(formula = Sleep_Duration ~ Age + Sleep_Quality + Physical_Activity +
    Stress_Level + Heart_Rate + Daily_Steps + Gender + Occupation +
    BMI_Category + SystolicBP + DiastolicBP + Sleep_Disorder,
    data = sleep)

Residuals:
     Min       1Q   Median       3Q      Max
-0.72133 -0.14300 -0.04099  0.13125  0.97206

Residual standard error: 0.244 on 350 degrees of freedom
Multiple R-squared:  0.9118,    Adjusted R-squared:  0.906
F-statistic: 157.3 on 23 and 350 DF,  p-value: < 2.2e-16
```

```
Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                     6.475e+00  1.151e+00   5.626 3.78e-08 ***
Age                             2.677e-02  6.385e-03   4.193 3.50e-05 ***
Sleep_Quality                   2.879e-01  5.597e-02   5.144 4.49e-07 ***
Physical_Activity               9.303e-03  1.551e-03   5.999 4.95e-09 ***
Stress_Level                   -1.627e-01  3.414e-02  -4.767 2.75e-06 ***
Heart_Rate                      3.253e-02  1.007e-02   3.231 0.001352 **
Daily_Steps                    -1.271e-04  2.171e-05  -5.853 1.11e-08 ***
GenderMale                     -2.146e-02  8.451e-02  -0.254 0.799732
OccupationDoctor                8.372e-01  8.547e-02   9.795  < 2e-16 ***
OccupationEngineer              7.789e-01  8.663e-02   8.991  < 2e-16 ***
OccupationLawyer                7.282e-01  9.885e-02   7.367 1.27e-12 ***
OccupationManager               9.962e-02  2.597e-01   0.384 0.701566
OccupationNurse                 2.283e-01  1.089e-01   2.096 0.036796 *
OccupationSales Representative  1.440e+00  2.349e-01   6.128 2.40e-09 ***
OccupationSalesperson           6.349e-01  1.079e-01   5.884 9.38e-09 ***
OccupationScientist             4.480e-01  1.691e-01   2.650 0.008418 **
OccupationSoftware Engineer     6.184e-01  1.478e-01   4.183 3.65e-05 ***
OccupationTeacher               2.792e-01  8.632e-02   3.234 0.001336 **
BMI_CategoryObese              -5.880e-01  1.921e-01  -3.061 0.002377 **
BMI_CategoryOverweight         -3.310e-01  9.771e-02  -3.388 0.000786 ***
SystolicBP                     -1.192e-01  1.591e-02  -7.491 5.62e-13 ***
DiastolicBP                     1.340e-01  2.173e-02   6.169 1.89e-09 ***
Sleep_DisorderNone             -9.748e-02  5.948e-02  -1.639 0.102174
Sleep_DisorderSleep Apnea      -4.719e-02  6.551e-02  -0.720 0.471825
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Variance Inflation Factors (VIF)

```r
full_model <- lm(Sleep_Duration ~ Age + Sleep_Quality +
                   Physical_Activity + Stress_Level +
                   Heart_Rate + Daily_Steps + Gender +
                   Occupation + BMI_Category + SystolicBP +
                   DiastolicBP + Sleep_Disorder,
                 data = sleep)
```

```r
> vif(full_model)
                       GVIF Df GVIF^(1/(2*Df))
Age                19.219673  1        4.384025
Sleep_Quality      28.128115  1        5.303595
Physical_Activity   6.539543  1        2.557253
Stress_Level       23.001854  1        4.796025
Heart_Rate         10.870145  1        3.296990
Daily_Steps         7.730518  1        2.780381
Gender             11.218893  1        3.349462
Occupation       2419.267736 10        1.476333
BMI_Category       74.750930  2        2.940385
SystolicBP         95.227925  1        9.758480
DiastolicBP       112.330917  1       10.598628
Sleep_Disorder     11.010121  2        1.821579
```

Full Model with all 12 predictors

```r
full_model <- lm(Sleep_Duration ~ Age + Sleep_Quality +
                   Physical_Activity + Stress_Level +
                   Heart_Rate + Daily_Steps + Gender +
                   Occupation + BMI_Category + SystolicBP +
                   Sleep_Disorder, data = sleep)
```

```r
> vif(full_model)
                       GVIF Df GVIF^(1/(2*Df))
Age                16.831587  1        4.102632
Sleep_Quality      28.019907  1        5.293383
Physical_Activity   6.471034  1        2.543823
Stress_Level       21.972472  1        4.687480
Heart_Rate         10.869956  1        3.296962
Daily_Steps         6.325794  1        2.515113
Gender             11.094874  1        3.330897
Occupation       1536.783726 10        1.443214
BMI_Category       53.495532  2        2.704453
SystolicBP         12.711108  1        3.565264
Sleep_Disorder     10.631461  2        1.805711
```

Model Without `DiastolicBP`

To make GVIFs comparable across dimensions, use GVIF^(1/(2*Df)), where Df is the number of coefficients in the subset.

Stepwise AIC Model Selection

```r
reduced_model <- stepAIC(fit0, direction = "forward",
                         scope=list(lower=fit0, upper=full_model))
```

```
Step:  AIC=-996.67
Sleep_Duration ~ Sleep_Quality + Occupation + Stress_Level +
    Physical_Activity + SystolicBP + Daily_Steps + Age + Sleep_Disorder +
    Heart_Rate + BMI_Category

         Df Sum of Sq    RSS     AIC
<none>                23.144 -996.67
+ Gender  1  0.049021 23.095 -995.46
```

- Starting from the model with no predictors (fit0)
- Iteratively adds the most contributive predictors, and stops when the improvement is no longer statistically significant.

ANOVA

```r
anova(reduced_model, full_model)
```

```
Analysis of Variance Table

Model 1: Sleep_Duration ~ Sleep_Quality + Occupation + Stress_Level +
    Physical_Activity + SystolicBP + Daily_Steps + Age + Sleep_Disorder +
    Heart_Rate + BMI_Category
Model 2: Sleep_Duration ~ Age + Sleep_Quality + Physical_Activity + Stress_Level +
    Heart_Rate + Daily_Steps + Gender + Occupation + BMI_Category +
    SystolicBP + Sleep_Disorder
  Res.Df    RSS Df Sum of Sq     F Pr(>F)
1    352 23.144
2    351 23.095  1  0.049021 0.745 0.3886
```

The ANOVA table confirms that **Gender**, the only variable that was dropped, did not contribute significantly to the model and because the p-value is much larger than 0.05, our reduced model is a better fit for our data

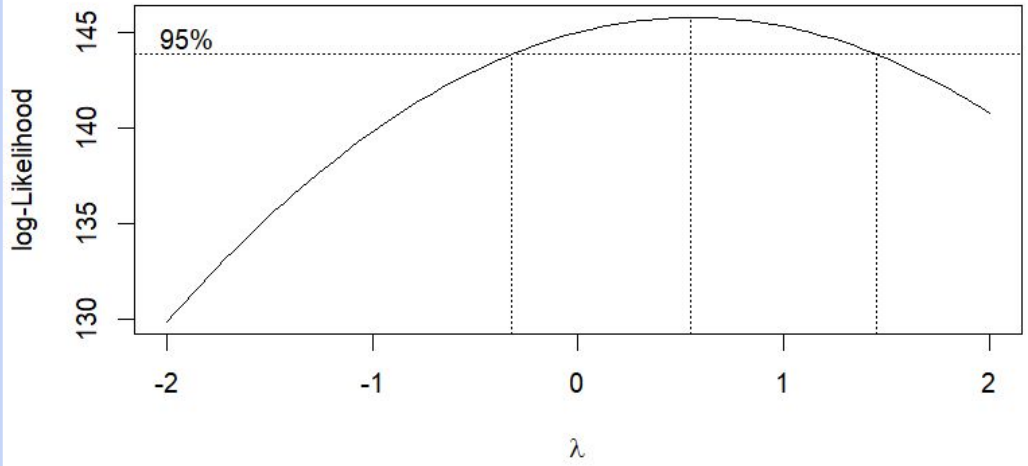
Lasso Model

```
12 x 1 sparse Matrix of class "dgCMatrix"
                              s0
Gender                 .
Age                    9.318538e-05
Occupation             .
Sleep_Quality          5.249059e-01
Physical_Activity      9.954789e-03
Stress_Level          -2.881551e-02
BMI_Category           .
systolic              -1.396510e-02
diastolic              .
Heart_Rate             8.485981e-03
Daily_Steps           -1.109375e-04
Sleep_Disorder         .
```

```
> lasso_reg$lambda.min
[1] 0.001584893
```

- The model effectively identifies key factors affecting sleep duration, which can help to improve sleep health.
- The use of Lasso regression helped simplify the model by eliminating irrelevant predictors.
- **Positive Coefficients**
  - Sleep_Quality, Physical_Activity, Heart_Rate
  - Indicate that increases in these predictors are associated with an increase in sleep duration.
- **Negative Coefficients**
  - Stress_Level, Systolic BP, Daily_Steps
  - Indicate that increases in these predictors are associated with a decrease in sleep duration.

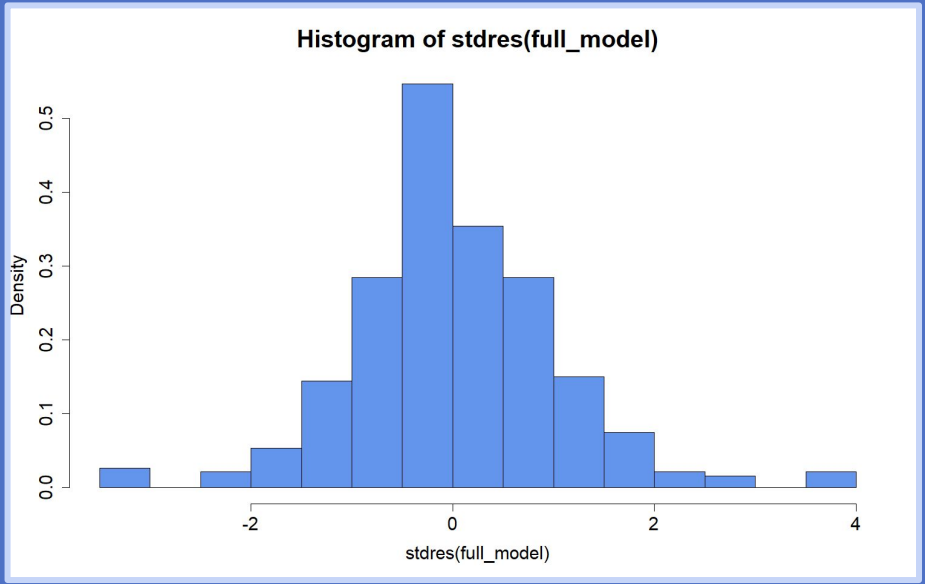
Transformation

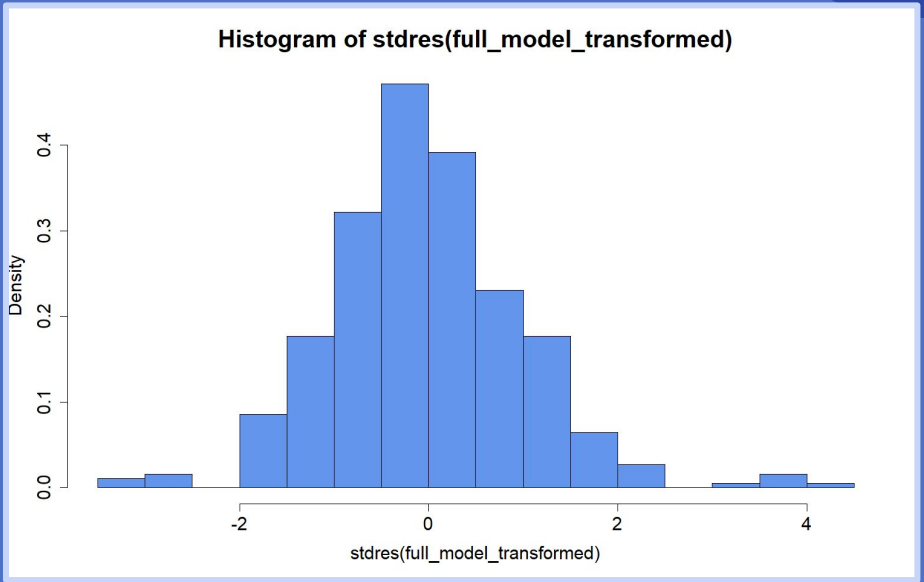


Curve Shape: The curve peaks around λ = 0.5, which corresponds to a square root transformation of the response variable

Normalization Effect: The square root transformation reduces the impact of extreme values or outliers

Residual Analysis after Transformation





Pre-Transformed → **Post-Transformed**

Through data transformation → normalize our distribution of residuals

2nd plot is better fit for normality

04

Conclusion

Results and Future Work

Results Analysis

- **Model Evaluation**
  - Reduced model selected based on ANOVA

- **Key Predictors and Their Impacts**
  - **Positive coefficients:** Sleep Quality, Physical Activity, and Heart Rate suggest these factors increase sleep duration

  - **Negative coefficients:** Stress Level, Systolic BP, and Daily Steps indicate these factors decrease sleep duration

- **Statistical Insights**
  - All key predictors are statistically significant with p-values < 0.05

Conclusions

- Model: `Sleep_Duration = 6.436 + Sleep_Quality + Occupation + Stress_Level + Physical_Activity + SystolicBP + Daily_Steps + Age + Sleep_Disorder + Heart_Rate + BMI_Category`

```
Multiple R-squared:  0.9022,
Adjusted R-squared:  0.8961
```

- Future Goals:
  - Greater sample size of ages, specifically under 20 and 60+
  - More diversity in occupations, can include students as well

```
Coefficients:
                                    Estimate
(Intercept)                        6.436e+00
Sleep_Quality                      2.881e-01
OccupationDoctor                   9.265e-01
OccupationEngineer                 7.635e-01
OccupationLawyer                   7.033e-01
OccupationManager                  6.295e-02
OccupationNurse                    4.597e-01
OccupationSales Representative     1.432e+00
OccupationSalesperson              5.452e-01
OccupationScientist                4.569e-01
OccupationSoftware Engineer        6.794e-01
OccupationTeacher                  2.618e-01
Stress_Level                      -2.171e-01
Physical_Activity                  9.845e-03
SystolicBP                        -2.968e-02
Daily_Steps                       -6.671e-05
Age                                1.670e-02
Sleep_DisorderNone                -1.577e-01
Sleep_DisorderSleep Apnea         -4.707e-02
Heart_Rate                         3.277e-02
BMI_CategoryObese                 -5.288e-01
BMI_CategoryOverweight            -4.556e-02
```

Thanks for not falling asleep!
Any Questions?